

MANUFACTURING TECHNOLOGY

Manufacturing is the transformation of materials into products to meet human needs and wants. Manufacturing technology courses provide students with opportunities to acquire knowledge and skills in the areas of mechanical engineering, robotics and control systems, computer-aided manufacturing, industrial maintenance, precision machining, welding, and sheet metal.

Students will use a broad range of tools and equipment while acquiring engineering, fabrication, and problem-solving skills through the design and fabrication of various projects. Manufacturing courses will also help students understand the effects of manufacturing technology on the environment and society.

Manufacturing courses are designed for students heading to an apprenticeship, to college or university, or directly to the workplace in their pursuit of careers such as tradesperson, technician, technologist, engineer, or any of the skilled entry positions.

The list of approved emphasis areas for manufacturing technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Manufacturing Technology, Grade 11

College Preparation

TMJ3C

This course enables students to develop knowledge and skills through hands-on, project-based learning. Students will acquire design, fabrication, and problem-solving skills while using tools and equipment such as lathes, mills, welders, computer-aided machines, robots, and control systems. Students may have opportunities to obtain industry-standard certification and training. Students will develop an awareness of environmental and societal issues related to manufacturing and will learn about pathways leading to careers in the industry.

Prerequisite: None

A. MANUFACTURING TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of how a design process is used in the planning and development of a manufacturing project;
- A2.** demonstrate an understanding of material conversion processes;
- A3.** demonstrate an understanding of various types of control systems used in manufacturing;
- A4.** apply relevant mathematical and scientific concepts and use appropriate forms of technical communication in the study of manufacturing technology.

SPECIFIC EXPECTATIONS

A1. Design Process

By the end of this course, students will:

- A1.1** describe and explain the importance of the steps in a design process (*e.g., analysing the situation, defining the need or problem, conducting research, generating solutions, communicating ideas, evaluating solutions, building a prototype, undertaking a cost analysis*) or other problem-solving process used to solve a variety of manufacturing technology challenges (*e.g., design and plan the fabrication of a robot or robotic work cell to specific design criteria and timelines*) (see pp. 21–23);
- A1.2** describe some examples of how the design process has been used in manufacturing to develop new products, or to improve existing products, to meet human needs or wants (*e.g., quality of automobiles improved through the use of robotics in the manufacturing process*);
- A1.3** explain why technological concepts (*e.g., aesthetics, control, environmental sustainability/stewardship, ergonomics, fabrication, function, innovation, material, mechanism, power and energy, structure, safety, systems*) are important considerations in the design process (see pp. 7–8);
- A1.4** use brainstorming strategies, web diagrams, and trial-and-error activities as part of the design process to develop the best possible solution to a manufacturing challenge.

A2. Material Conversion

By the end of this course, students will:

- A2.1** describe and demonstrate the correct use of a variety of processes for joining materials (*e.g., welding, bonding, fastening*);
- A2.2** describe and demonstrate the correct use of a variety of processes for cutting materials (*e.g., using oxy-acetylene torches, saws, shears, a plasma cutter, an engine lathe, a milling machine*);
- A2.3** describe and demonstrate the correct use of a variety of processes for forming materials, using various tools and equipment (*e.g., hammers; brake press; rollers; forging, casting, and moulding equipment*);
- A2.4** describe material conversions such as the separation process (*e.g., converting a material's size and shape by removing excess material*); the addition process (*e.g., combining materials to achieve enhanced qualities, as in alloys*); the process of making changes to contours (*e.g., assembling materials by such means as gluing, mixing, fastening, bonding, welding*); and the process of changing the properties (*e.g., thermal, chemical, mechanical, physical*) of solid materials;
- A2.5** demonstrate a working knowledge of converting the structure of a material, using a variety of methods (*e.g., heat treating, annealing, steaming, shrinking, tempering*).

A3. Control Systems

By the end of this course, students will:

- A3.1** describe various manufacturing control systems (*e.g., inventory/purchasing control, quality control*);
- A3.2** explain the use of control system technologies (*e.g., electronic – programmable logic controller [PLC]; hydraulic; mechanical; pneumatic*) in a production process or project;
- A3.3** operate a basic control system (*e.g., electronic, pneumatic, hydraulic, mechanical*) to control a robot or component in completing a simple task.

A4. Mathematics, Science, and Communication Skills

By the end of this course, students will:

- A4.1** demonstrate a working knowledge of various mathematical formulas (*e.g., Pythagorean theorem, formulas for calculating volume and surface area*) and applications (*e.g., tap drill sizing, calculation of machine speeds and feeds, weight calculations*) that are commonly used in manufacturing;
- A4.2** apply mathematical concepts (*e.g., linear systems; integers; decimals and fractions; order of operations*) and skills in performing a variety of tasks required within the context of manufacturing design and production (*e.g., angle calculations; calculation of perimeter, volume, and area; percent/decimal/fraction conversions; US customary/British imperial and metric unit conversions*);
- A4.3** apply mathematical analysis skills as they relate to production cost and quality control when manufacturing a product (*e.g., conduct a cost analysis of the final product or process, produce statistical process control charts*);
- A4.4** apply scientific laws, theories, and concepts related to the design and production areas of manufacturing (*e.g., laws/theories: Newton's laws of motion, basic electrical theory; concepts: simple machines, mechanical advantage*);
- A4.5** identify the properties (*e.g., magnetic, physical, mechanical, thermal, electrical*) of natural and manufactured solid materials;
- A4.6** describe how the physical and mechanical properties of solid materials (*e.g., physical: density, appearance, porosity, surface texture; mechanical: hardness, ductility, elasticity, brittleness, compression, fatigue, shear, tensility*) create advantages or disadvantages that affect their suitability for use in manufacturing;
- A4.7** use technical language correctly and appropriately (*e.g., in reports and presentations*) to communicate information related to manufacturing technology;
- A4.8** use appropriate forms of communication (*e.g., oral and written reports*) and supporting documentation (*e.g., statistical graphing; sketches; scale drawings; perspectives; orthographic, isometric, and oblique views; pattern and development projections; shop, assembly, and detail drawings*) to convey ideas and processes and to show materials and specifications.

B. MANUFACTURING TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** demonstrate the ability to interpret and prepare technical drawings and develop process plans;
- B2.** demonstrate a working knowledge of the characteristics of various materials and the proper selection of materials for the manufacture of a product;
- B3.** demonstrate a working knowledge of various metrology tools used to measure, lay out, and inspect products;
- B4.** use tools, equipment, and machine processes safely and correctly in the manufacture of a product.

SPECIFIC EXPECTATIONS

B1. Technical Drawings and Process Plans

By the end of this course, students will:

- B1.1** demonstrate a working knowledge of the stages of the manufacturing process (*e.g., produce an order of operation chart, develop a bill of material for a project, undertake a cost analysis, develop a process plan*);
- B1.2** use appropriate language and terminology (*e.g., in oral and written reports*) when developing process plans;
- B1.3** convert drawing dimensions (*e.g., US customary/British imperial units to metric units, fractions to decimals*) using the appropriate chart, table, or formula;
- B1.4** prepare, using conventional means or computer-aided design (CAD), detailed working drawings and/or assembly drawings or rough sketches (*e.g., section views, isometric views*), as appropriate to the task, that include the measurements and symbols required to complete a project (*e.g., dimensions, specifications, tolerances, surface finish symbols, weldments*).

B2. Material Characteristics and Selection

By the end of this course, students will:

- B2.1** identify the factors that affect material selection and suitability for the manufacture of a product (*e.g., hardness, softness, malleability, mass, durability, shape, size, finish*);

- B2.2** select appropriate materials for a project based on their properties and characteristics (*e.g., physical, mechanical, thermal, chemical, electrical, magnetic, optical, acoustical*) and the project's design criteria (*e.g., strength, finish, customer specifications, quality control processes, working environment*);

- B2.3** use the proper procedures to prepare materials for production (*e.g., measuring, marking, cutting to rough length, grinding, cleaning, deburring*).

B3. Metrology Tools

By the end of this course, students will:

- B3.1** demonstrate the correct use of various measuring instruments (*e.g., scales, vernier calipers, micrometers, gauges*) to make measurements in both metric and US customary/British imperial units;
- B3.2** apply principles of dimensional metrology (*e.g., precision measurement, tolerancing for interchangeable manufacturing*) to manufacturing processes when producing a product;
- B3.3** demonstrate a working knowledge of the proper tools required to lay out a product precisely (*e.g., height gauge, surface gauge, trammel points, protractor, scribe, straight edge*);
- B3.4** use metrology tools to inspect and evaluate products for quality control purposes (*e.g., for inspection reports, destructive and non-destructive testing*).

B4. Tools, Equipment, and Machine Processes

By the end of this course, students will:

- B4.1** demonstrate the use of appropriate bench work techniques to lay out, fit, and assemble workpieces;
- B4.2** use appropriate procedures (*e.g., correct machine set-up, operational safety procedures*) when setting up, maintaining, using, and storing tools and equipment used in manufacturing and production processes;

B4.3 demonstrate the correct selection and use of appropriate tools and equipment (*e.g., wrenches, electric drills, grinders, engine lathe, milling machine*) for specific manufacturing tasks;

B4.4 demonstrate the use of various bonding, combining, and/or assembly techniques (*e.g., shielded metal arc welding [SMAW], gas metal arc welding [GMAW], gas tungsten arc welding [GTAW], oxy-acetylene welding, brazing, soldering, use of appropriate fasteners, bonding*) to complete projects.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** demonstrate an understanding of ways in which the manufacturing industry affects the environment;
- C2.** demonstrate an understanding of ways in which the manufacturing industry affects society.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the benefits of using environmentally friendly products in the manufacturing process (*e.g., the benefits of water-based versus solvent-based adhesives*);
- C1.2** identify a variety of alternative fuels (*e.g., biodiesel, ethanol, hydrogen, electric power in hybrid systems*) and energy sources (*e.g., wind power, solar power, waste-to-energy*) and explain how use of these fuels and energy sources can reduce the environmental impact of the manufacturing industry;
- C1.3** explain how various sources of power generation used in manufacturing (*e.g., coal, nuclear, solar, wind, hydrogen fuel cell, tidal, geothermal*) affect the environment (*e.g., construction of large hydroelectric dams can affect animal habitats and patterns of behaviour*);
- C1.4** describe environmentally responsible practices that can be followed during the design and manufacture of a product (*e.g., minimize waste, consider using renewable or recyclable materials, design and manufacture products that last or can be repaired as opposed to throw-away products, use processes that have minimal impact on workers and the local environment*);
- C1.5** demonstrate the use of proper techniques for the disposal of waste products.

C2. Technology and Society

By the end of this course, students will:

- C2.1** explain how the manufacturing industry affects the local and provincial economy (*e.g., with respect to job creation, standards of living, sustainability and conservation of the environment, impact on First Nation communities*);
- C2.2** describe how the manufacturing industry responds to changes in the global economy (*e.g., fluctuating currencies, trade agreements, fluctuating demand in other countries*) and how this response affects the local community and/or the province as a whole (*e.g., in terms of economic opportunities or risks for the local and/or provincial population*);
- C2.3** explain how ergonomics and a good manufacturing work environment can affect people's daily lives (*e.g., by providing long-term health benefits, creating job efficiencies*).

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and compliance with health and safety legislation, standards, and practices related to the manufacturing industry;
- D2.** demonstrate an understanding of career opportunities in the manufacturing industry and the education, training, and certification required for these careers.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** identify and explain the importance of the specific components of legislation and standards related to workplace safety in the manufacturing industry (e.g., *Occupational Health and Safety Act [OHSA]*, *Workplace Hazardous Materials Information System [WHMIS]*) and the specific responsibilities of the relevant oversight/regulatory organizations (e.g., *Workplace Safety and Insurance Board [WSIB]*, *Industrial Accident Prevention Association [IAPA]*, *Construction Safety Association of Ontario [CSAO]*);
- D1.2** demonstrate good housekeeping practices in the work environment (e.g., *cleaning up spills and leaks, keeping areas clean and clear of obstructions, properly organizing tools and equipment*);
- D1.3** handle materials safely and appropriately in compliance with the information included in the Material Safety Data Sheets (MSDS) from the Workplace Hazardous Materials Information System (WHMIS);
- D1.4** demonstrate an understanding of procedures to ensure safe and productive work practices in the manufacturing workplace (e.g., *perform safety inspections and audits that include ergonomic considerations related to workshop layout and set-up, material handling, ease of movement, lighting, workstation set-up*);
- D1.5** demonstrate the safe use of tools and equipment in compliance with safety manuals, instructions, and institutional requirements;

D1.6 use proper ventilation systems to control air quality (e.g., *to minimize the effects of welding fumes, plastic off-gassing, cutting-fluid misting, heat treating of metal*);

D1.7 use protective clothing and equipment as required to ensure their own and others' safety in the work environment.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify employment opportunities available in the manufacturing industry locally, provincially, and nationally;
- D2.2** identify manufacturing-related college programs and apprenticeships that are available in the manufacturing industry prior to or immediately following graduation;
- D2.3** identify opportunities available locally to obtain certifications and/or training that it would be beneficial to have if they are pursuing a career in manufacturing (e.g., *cardiopulmonary resuscitation [CPR]*, *Standard First Aid*, *fall protection training*);
- D2.4** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the manufacturing industry (e.g., *mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

D2.5 demonstrate an understanding of and apply the Essential Skills that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., decision making, job task planning and organizing, problem solving*);

D2.6 demonstrate an understanding of and apply the work habits that are important for success in the manufacturing industry, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, initiative, working independently*);

D2.7 maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in manufacturing technology (*e.g., Passport to Safety certificate, Ontario Skills Passport Work Plan, photographs of a project, technical drawings, reports, assignments*), and explain why having a current portfolio is important for career development and advancement.